

**Started on** Tuesday, 24 January 2023, 9:26 AM

**State** Finished

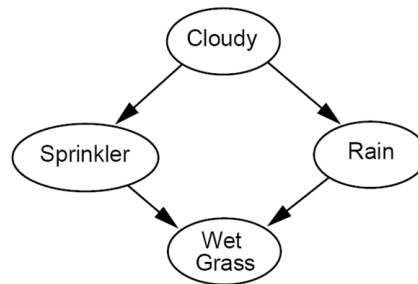
**Completed on** Tuesday, 24 January 2023, 10:55 AM

**Time taken** 1 hour 29 mins

**Grade** 39.33 out of 59.00 (66.67%)

Information

Consider the following Bayesian network that models the relation between weather, the state of the lawn in front of your house, and an automatic sprinkler you have installed there, with four Boolean variables abbreviated as  $C$ ,  $S$ ,  $R$ ,  $W$ .



Information

Which of the following statements are correct, given the structure of this model?

Question 1

Correct

Mark 1.00 out of 1.00

$(S \perp R)$

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question 2

Correct

Mark 1.00 out of 1.00

$$(R \perp W \mid C, S)$$

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question 3

Correct

Mark 1.00 out of 1.00

$$P(W \mid C) = P(C \mid W)$$

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question 4

Incorrect

Mark 0.00 out of 1.00

$$P(C \mid R, W) = P(C \mid R)$$

Select one:

- ☒ True ✗
- ☐ False

The correct answer is 'False'.

Question 5

Correct

Mark 1.00 out of 1.00

$$(W \perp C \mid S, R)$$

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.



## Question 6

Correct

Mark 1.00 out of 1.00

$$P(R | S) = P(R)$$

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

## Question 7

Incorrect

Mark 0.00 out of 1.00

$$(R \perp W | C)$$

Select one:

- ☒ True ✗
- ☐ False

The correct answer is 'False'.

## Question 8

Correct

Mark 1.00 out of 1.00

Suppose we represented the additional information that wet grass directly tells us something about whether the sky is cloudy by adding an edge  $W \rightarrow C$  to this model. Which one of the following statements would then be true about the new model?

- ☐ The CPD table for variable  $W$  now has 8 rows
- ☐ The CPD table for variable  $C$  now has 4 rows
- ☐ The CPD table for variable  $C$  now has 2 rows
- ☒ This is not a valid Bayes net structure ✓
- ☐ The CPD table for variable  $C$  now has 3 rows

Your answer is correct.

The correct answer is:

This is not a valid Bayes net structure

## Information

Now assume we remove all edges from this network, leaving only four unconnected variables. Which statements are true about this structure:



Question **9**

Incorrect

Mark 0.00 out of 1.00

This model would not permit any probabilistic inference.

Select one:

- ☒ True 
- ☐ False

The correct answer is 'False'.

Question **10**

Correct

Mark 1.00 out of 1.00

This would be an extremely high-bias model structure.

Select one:

- ☒ True 
- ☐ False

The correct answer is 'True'.

Question **11**

Incorrect

Mark 0.00 out of 1.00

This model would only permit evidential reasoning.

Select one:

- ☒ True 
- ☐ False

The correct answer is 'False'.

Question **12**

Correct

Mark 1.00 out of 1.00

This is not a valid structure for a Bayesian network.

Select one:

- ☐ True
- ☒ False 

The correct answer is 'False'.



Question **13**

Correct

Mark 1.00 out of 1.00

This model would only require 4 numbers to be fully parametrized.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Information

Suppose we inverted the direction of all arrows in the original model and estimated Maximum Likelihood parameters for the model from some training dataset. Which of the following statements about the resulting model are correct?

Question **14**

Incorrect

Mark 0.00 out of 1.00

The model contradicts reality because it implies that rain and the sprinkler can cause a cloudy sky.

Select one:

- ☒ True ✗
- ☐ False

The correct answer is 'False'.

Question **15**

Correct

Mark 1.00 out of 1.00

The model would represent the same joint distribution as the original one.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.



Question 16

Correct

Mark 1.00 out of 1.00

The model represents the same independencies as the original one.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question 17

Correct

Mark 1.00 out of 1.00

The model supports only evidential reasoning.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Information

Now consider the following distributions over these four variables, represented as multidimensional tables (forget about the network structure). For each of the following tables, give the number of dimensions, the total number of values stored in these, and the number of non-redundant (independent) ones:

Question 18

Partially correct

Mark 0.67 out of 1.00

$P(C, W)$

Dimensions:  ✓

Entries:  ✓

Non-Redundant Entries:  ✗



Question **19**

Partially correct

Mark 0.67 out of 1.00

$$P(W)$$

Dimensions:  ✓

Entries:  ✓

Non-Redundant Entries:  ✗

Question **20**

Incorrect

Mark 0.00 out of 1.00

$$P(C, S, R \mid W)$$

Dimensions:  ✗

Entries:  ✗

Non-Redundant Entries:  ✗

Question **21**

Partially correct

Mark 0.67 out of 1.00

$$P(W \mid C)$$

Dimensions:  ✓

Entries:  ✗

Non-Redundant Entries:  ✓



Question **22**

Partially correct

Mark 0.67 out of 1.00

$$P(R, S)$$

Dimensions:  ✓

Entries:  ✓

Non-Redundant Entries:  ✗

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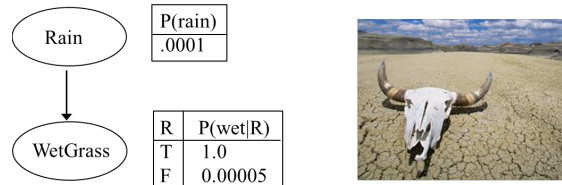
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Information

Consider again the Death Valley model from the lecture slides:



Question 23

Correct

Mark 1.00 out of 1.00

Consider a set  $\mathcal{D}$  of 100 samples drawn from this network via Forward Sampling. What is the probability estimate  $P(\text{rain} \mid \neg \text{wet})$  that you will obtain from  $\mathcal{D}$  with Rejection Sampling?

Answer: 0 ✓

The correct answer is: 0

Question 24

Incorrect

Mark 0.00 out of 1.00

Consider the probabilistic query  $P(\text{WetGrass} \mid \neg \text{rain})$ . When you use Likelihood Weighting to answer this, what will be the weight given to all the samples?

Answer: 0.99989995 ✗

The correct answer is: 0.9999

Question 25

Correct

Mark 1.00 out of 1.00

Can Forward Sampling produce a sample  $(\text{rain}, \neg \text{wet})$  from this model?

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **26**

Correct

Mark 1.00 out of 1.00

Can  $P(\text{rain} \mid \text{wet})$  be calculated via Bayes' rules, using only numbers from this network (including, of course, the redundant ones we did not explicitly list in the tables)?

Select one:

☒ True ✓

☐ False

The correct answer is 'True'.

Question **27**

Correct

Mark 1.00 out of 1.00

Compute the *exact answer* (all decimal places) to the query  $P(\text{WetGrass})$ .

true:  ✓

false:  ✓

Question **28**

Correct

Mark 1.00 out of 1.00

Compute the *exact answer* (all decimal places) to the query  $P(\text{Rain})$ .

true:  ✓

false:  ✓

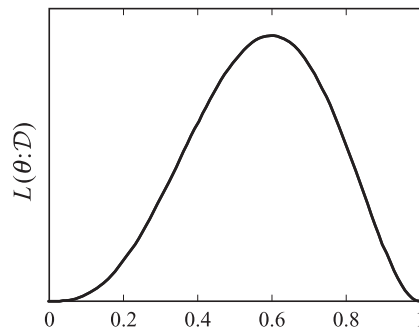
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Information

Consider the example, from the lecture slides, of learning the parameter  $\theta = P(\text{heads})$  for our thumbtack, where we had the following likelihood function  $L(\theta, \mathcal{D})$  based on a set  $\mathcal{D}$  of  $N = 5$  observations, of which  $k = 3$  were heads, and  $(N - k) = 2$  were tails:



Which of the following statements are true (about this specific likelihood function, relating to this specific dataset):

Question 29

Incorrect

Mark 0.00 out of 1.00

$$L(0.6 : \mathcal{D}) = 3/5$$

Select one:

- ☒ True ✖
- ☐ False

The correct answer is 'False'.

Question 30

Correct

Mark 1.00 out of 1.00

$$L(0.6 : \mathcal{D}) = 1$$

Select one:

- ☐ True
- ☒ False ✔

The correct answer is 'False'.

Question **31**

Correct

Mark 1.00 out of 1.00

$L(0.5 : \mathcal{D})$  is the probability that  $\theta = 0.5$  would generate a dataset with 3 heads if we flipped the thumbtack 5 times.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **32**

Incorrect

Mark 0.00 out of 1.00

$L(0.6 : \mathcal{D})$  is the probability that  $\theta = 0.6$  would generate a dataset with 300 heads if we flipped the thumbtack 500 times.

Select one:

- ☒ True ✗
- ☐ False

The correct answer is 'False'.

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Information

Which of the following statements about Structure Learning from a dataset  $\mathcal{D}$  are correct?

Question **33**

Correct

Mark 1.00 out of 1.00

If we redefined the model dimension ( $Dim$ ) in the BIC score to consider not just the *non-redundant parameters* but the fully written-out CPDs, that could lead it to learn different models.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **34**

Correct

Mark 1.00 out of 1.00

The Maximum Likelihood Score  $\text{score}_L(\mathcal{G} : \mathcal{D})$  for a graph structure  $\mathcal{G}$  that is fully connected (i.e., has the maximum possible number of edges) is 1.0.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **35**

Correct

Mark 1.00 out of 1.00

The BIC score punishes models with a high likelihood.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **36**

Incorrect

Mark 0.00 out of 1.00

Redefining the model dimension (*Dim*) in the BIC score to consider not just the *non-redundant* parameters but the fully written-out CPDs, would increase the learner's bias.

Select one:

- ☐ True
- ☒ False ✖

The correct answer is 'True'.

Question **37**

Correct

Mark 1.00 out of 1.00

The BIC score uses the likelihood as a measure of fit on the data.

Select one:

- ☒ True ✔
- ☐ False

The correct answer is 'True'.

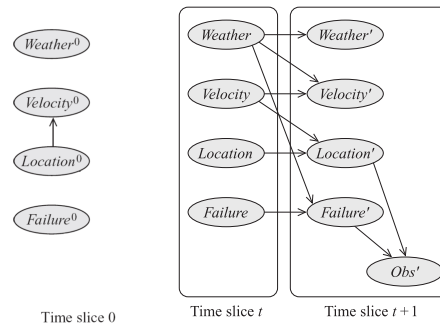
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Information

Assume we have the following state-observation model of a moving car (taken from the lecture slides), where variable Obs is the only observable one. (Only the structure of the model is shown, not the actual CPD tables; also: assume that all variables are discrete).



Information

Which of the following statements about this S-O model and its parametrization are correct?

Question 38

Correct

Mark 1.00 out of 1.00

The Initial State Model  $P(\mathbf{S}^{(0)})$  is a vector with 4 numbers.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question 39

Correct

Mark 1.00 out of 1.00

The State Transition Model is defined by 8 (eight) CPD tables.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **40**

Incorrect

Mark 0.00 out of 1.00

The Observation Model is one CPD table.

Select one:

- ☐ True
- ☒ False **✖**

The correct answer is 'True'.

Question **41**

Correct

Mark 1.00 out of 1.00

The variable **Weather** could be dropped entirely, because it is not observable.

Select one:

- ☐ True
- ☒ False **✔**

The correct answer is 'False'.

Question **42**

Correct

Mark 1.00 out of 1.00

The State Transition Model is defined by 5 (five) CPD tables.

Select one:

- ☐ True
- ☒ False **✔**

The correct answer is 'False'.

Question **43**

Incorrect

Mark 0.00 out of 1.00

The State Transition Model is defined by 9 (nine) CPD tables.

Select one:

- ☒ True **✖**
- ☐ False

The correct answer is 'False'.



Question **44**

Correct

Mark 1.00 out of 1.00

If we added a link from Failure to Obs', this would not be a valid S-O model.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Information

Consider the tasks of Filtering, Smoothing, and Prediction as defined in the lecture slides. Which of the following statements are true (given the above S-O model):

Question **45**

Incorrect

Mark 0.00 out of 1.00

The series of state distributions  $(P(S^{(t+k)} \mid \mathrm{obs}^{(1:t)}))$  in Prediction forms a Markov Chain.

Select one:

- ☐ True
- ☒ False ✗

The correct answer is 'True'.

Question **46**

Correct

Mark 1.00 out of 1.00

The forward message computed by the filtering algorithm is a vector of 4 values.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question 47

Correct

Mark 1.00 out of 1.00

The value of  $\mathbb{P}(\text{obs}^{(1)})$  has no influence on the filtered distribution  $\mathbb{P}(S^{(t)} \mid \mathbb{P}(\text{obs}^{(1:t)}))$  for  $t > 2$ .

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question 48

Incorrect

Mark 0.00 out of 1.00

For all  $t < T$ ,  $\mathbb{P}(S^{(t)} \mid \mathbb{P}(\text{obs}^{(1:t)})) \neq \mathbb{P}(S^{(t)} \mid \mathbb{P}(\text{obs}^{(1:T)}))$ .

Select one:

- ☒ True ✗
- ☐ False

The correct answer is 'False'.

Question 49

Correct

Mark 1.00 out of 1.00

Calculating  $\mathbb{P}(S^{(t+k)} \mid \mathbb{P}(\text{obs}^{(1:t)}))$  (prediction) is identical to Filtering with observations  $\mathbb{P}(\text{obs}^{(t+k)} = 0)$ .

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question 50

Correct

Mark 1.00 out of 1.00

Smoothing can change all state distributions (compared to the filtered distributions) except the last one (at  $T$ ).

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

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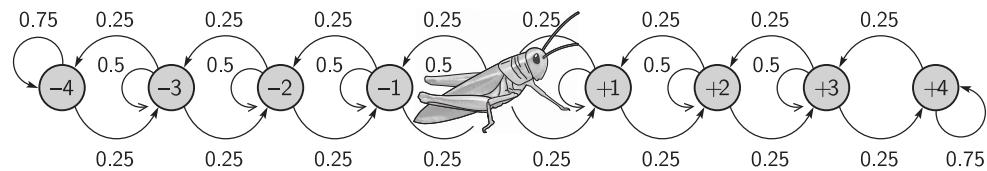
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Information

Consider again the Grasshopper Markov Chain from Chapter 4b, with a state variable

$S = \{-4, -3, -2, -1, 0, +1, +2, +3, +4\}$  and with the grasshopper known to start in position 0:

$P(S^{(0)}) = [0, 0, 0, 0, 1, 0, 0, 0, 0]$ :



Let's turn this into a Hidden Markov Model by assuming that we cannot see the grasshopper, but we hear it chirping (observation variable  $\text{Chirp}(C)$ ), with the sound coming from the left, the center, or the right ( $\text{Val}(C) = \{l, c, r\}$ ).

Suppose we have the following observation model:

| $S$ | $P(C   S)$ |     |     |
|-----|------------|-----|-----|
|     | $l$        | $c$ | $r$ |
| -4  | 1          | 0   | 0   |
| -3  | 1          | 0   | 0   |
| -2  | 1          | 0   | 0   |
| -1  | 1          | 0   | 0   |
| 0   | 0          | 1   | 0   |
| +1  | 0          | 0   | 1   |
| +2  | 0          | 0   | 1   |
| +3  | 0          | 0   | 1   |
| +4  | 0          | 0   | 1   |

Finally, suppose we have an observation sequence of Chirp observations of length  $T$ . Which of the following sentences are correct:

Question 51

Incorrect

Mark 0.00 out of 1.00

This model is equivalent to a Markov chain where the grasshopper is directly visible.

Select one:

- ☒ True ✗
- ☐ False

The correct answer is 'False'.

Question **52**

Incorrect

Mark 0.00 out of 1.00

The backward message  $b^{(1:T)} \approx [1/9, 1/9, 1/9, 1/9, 1/9, 1/9, 1/9, 1/9, 1/9]^\top$

Select one:

- ☒ True ✖
- ☐ False

The correct answer is 'False'.

Question **53**

Incorrect

Mark 0.00 out of 1.00

The forward message  $f^{(1:0)} = [0, 0, 0, 0, 1, 0, 0, 0, 0]^\top$

Select one:

- ☐ True
- ☒ False ✖

The correct answer is 'True'.

Question **54**

Correct

Mark 1.00 out of 1.00

The forward message in Filtering will always be a vector with one 1 and eight zeroes.

Select one:

- ☐ True
- ☒ False ✔

The correct answer is 'False'.

Question **55**

Incorrect

Mark 0.00 out of 1.00

The backward message will always sum up to 1.0

Select one:

- ☒ True ✖
- ☐ False

The correct answer is 'False'.

Question 56

Correct

Mark 1.00 out of 1.00

Decoding: Given the observation sequence  $(r, r)$ , what is the correct  $\hat{s}^{(0:2)} = \arg \max_{\mathbf{s}^{(0:2)}} P(\mathbf{s}^{(0:2)} | (r, r))$ ?

- ☐  $(+4, +4)$
- ☐  $(0, +1, +2)$
- ☐  $(-4, 0, +4)$
- ☐  $(-1, 0, +1)$
- ☐  $\backslash (0, +1, 0) \backslash$
- ☒  $\backslash (0, +1, +1) \backslash$  ✓
- ☐  $\backslash (+1, +1) \backslash$
- ☐  $\backslash (+1, +2) \backslash$

Your answer is correct.

The correct answer is:  $\backslash (0, +1, +1) \backslash$

Question 57

Correct

Mark 1.00 out of 1.00

Decoding: Given the observation sequence  $\backslash (r, r) \backslash$ , how many state sequences would we have to sum over if we solved the Decoding problem simply by evaluating all state sequences?

Answer:  ✓

The correct answer is: 729

Question 58

Partially correct

Mark 0.67 out of 1.00

Decoding: Give  $\backslash v^{\{1:0\}} \backslash$  for the above HMM model:

$\backslash v^{\{1:0\}} = \backslash$   ✓  ✓  ✓  ✗  ✗  ✗  ✓  
 ✓  ✓  ✓  ✓

Question **59**

Correct

Mark 1.00 out of 1.00

What is the likelihood of the above HMM w.r.t. the observation sequence  $\{c, c\}$ ?

$P(c, c) =$

- ☐ 1/9
- ☐ 1
- ☐ 0
- ☐ 8/9
- ☐ 0.5
- ☐ 1.11
- ☒ 0.25 ✓

Your answer is correct.

The correct answer is:  
0.25

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